

```
$ parallel echo {} ::: /tmp/stdio.h
/tmp/stdio
```

The output of a GNU Parallel command may not necessarily be in the order in which the input arguments are listed. For example:

```
$ parallel sleep {} \; echo {} ::: 5 2 1 4 3
1
2
4
3
5
```

If you wish to enforce the order of execution, use the '-k' option, as shown below:

```
$ parallel -k sleep {} \; echo {} ::: 5 2 1 4 3
5
2
1
4
3
```

A test script, for example, may need to be run 'N' times for the same argument. This can be accomplished with the following code:

```
$ seq 10 | parallel -n0 echo "Hello, World"
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
Hello, World
```

The '-n' option represents the maximum number of arguments in the command line.

The commands that will get executed by GNU Parallel can be observed with the '--dry-run' option, as illustrated below:

```
$ parallel --dry-run -k sleep {} \; echo {} ::: 5 2 1 4 3
sleep 5; echo 5
sleep 2; echo 2
sleep 1; echo 1
sleep 4; echo 4
sleep 3; echo 3
```

The '-eta' option will give an estimate of the time it will

take to complete a job:

```
$ parallel --eta -k sleep {} \; echo {} ::: 5 2 1 4 3
```

```
Computers / CPU cores / Max jobs to run
1:local / 4 / 4
```

```
Computer:jobs running/jobs completed/%of started jobs/Average
seconds to complete
ETA: 5s 1left 1500avg local:1/4/100%/1.0s
2
1
4
3
ETA: 1s 0left 1.00avg local:0/5/100%/1.0s
```

Suppose you have a large number of log files that you wish to zip and archive, you can run the gzip command in Parallel, as shown below:

```
$ parallel gzip ::: *.log
```

To unzip them all, you can use the following command:

```
$ parallel gunzip ::: *.gz
```

The 'convert' command is useful to transform image files. High resolution images can be scaled to a lower resolution using the following command:

```
$ convert -resize 512x384 file.jpg file_web.jpg
```

If you have a large number of files that you wish to resize, you can parallelise the task, as shown below:

```
$ find . -name '*.jpg' | parallel convert -resize 512x384 {}
{}_web.jpg
```

GNU Parallel with wget can help in parallel downloads of large Linux kernel releases, as shown below:

```
$ parallel wget ::: \
www.kernel.org/pub/linux/kernel/v3.x/linux-3.11.tar.xz
www.kernel.org/pub/linux/kernel/v3.x/linux-3.10.10.tar.xz
```

The URLs can also be stored in a text file ("input.txt"), and passed as an argument to Parallel:

```
$ parallel -a input.txt wget
```

The file 'input.txt' contains:

```
https://www.kernel.org/pub/linux/kernel/v3.x/linux-3.11.tar.xz
https://www.kernel.org/pub/linux/kernel/v3.x/linux-3.10.10.tar.xz
```